Selection Models for Publication Bias

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Introduction

Selection models address publication bias more rigorously than trim-and-fill or simple regression-based asymmetry tests. They posit an explicit weight function describing how the probability of publication depends on a study's p-value or effect magnitude, then re-estimate the pooled effect under that assumed selection mechanism to recover a bias-adjusted estimate. Where trim-and-fill is descriptive and triangular in its assumption, selection models are formally model-based and yield likelihood-ratio tests of the selection hypothesis itself. The cost is stronger parametric assumptions about the form of the selection process, which means selection models are most useful when the meta-analysis is large enough that the selection function can be reasonably identified from the data.

Prerequisites

A working understanding of publication bias as a selection process, maximum-likelihood estimation, and the logic of inverse-probability weighting on a sampled population.

Theory

Given observed effects y_i with standard errors SE_i from published studies, the observed density is

$$p_{\mathrm{obs}}(y) = \frac{w(y, \mathrm{SE})\, p(y)}{\int w(y, \mathrm{SE})\, p(y)\, dy},$$

where $p(y)$ is the population density of true effects and $w(\cdot)$ is a selection weight function — typically a step function in the one-sided p-value, with selection probability dropping at conventional thresholds (0.025, 0.05, 0.10). Maximum-likelihood estimation of the joint (μ, τ^2, w) parameters then inverts the selection mechanism to recover the bias-adjusted pooled effect, and a likelihood-ratio test against the null of $w(\cdot) \equiv 1$ formally tests for selection.

Assumptions

The functional form of the selection weight is correctly specified (or at least a reasonable approximation), the meta-analysis contains enough studies to identify the selection parameters (typically 15–20+), and the studies are exchangeable conditional on their effect size and standard error.

R Implementation

```r
library(metafor)

set.seed(2026)
k <- 20
yi <- rnorm(k, 0.3, 0.15)
vi <- 1 / sample(40:250, k, replace = TRUE)

res <- rma(yi, vi, method = "REML")
```

```r
sel <- selmodel(res, type = "stepfun",
                steps = c(0.025, 0.10, 0.50, 1.0),
                alternative = "greater")
sel
```

Output & Results

`selmodel()` returns the bias-adjusted pooled estimate, its standard error and confidence interval, the estimated selection weights at each step, and a likelihood-ratio test against the null of no selection. Comparing the adjusted and unadjusted estimates side by side quantifies how much the conclusion shifts under the assumed selection mechanism.

Interpretation

A reporting sentence: "A step-function selection model with thresholds at $p = 0.025, 0.10, 0.50$ provided strong evidence for selective publication (likelihood-ratio test vs no-selection, $\chi^2 = 13.1$, df = 3, $p = 0.004$). The bias-adjusted standardised mean difference was 0.19 (95 % CI 0.05 to 0.33), substantially attenuated from the unadjusted estimate of 0.32 (0.21 to 0.43). The result suggests publication bias inflates the unadjusted pooled effect by roughly 40 %." Always report both estimates and the LR test.

Practical Tips

- Selection models require at least 15–20 studies for stable estimates; below this threshold the selection function is poorly identified and confidence intervals become uninformatively wide.
- Try several weight-function structures (different step thresholds, smooth-monotone forms) as sensitivity analyses; the bias-adjusted estimate can be quite sensitive to the assumed selection mechanism, and convergent results across forms are reassuring.
- Always report both the unadjusted and the selection-adjusted estimates, plus the LR test for selection; the unadjusted estimate remains the primary, and the selection-adjusted estimate is sensitivity.
- Modern alternatives — PET-PEESE regression-based correction, p-uniform and p-curve, and Vevea–Hedges weight-function models — are useful when standard step-function selection models fit poorly or fail to converge.
- Bayesian selection models with informative priors on the weight parameters (e.g., in `RoBMA` and related packages) are a stable alternative when frequentist MLE is unstable.
- Selection models adjust only for the selection mechanism they are told to assume; other forms of bias (selective outcome reporting, dependent effect-size selection) require separate handling.

R Packages Used

`metafor::selmodel()` for the canonical step-function and smooth selection-model implementations; `weightr` for Vevea–Hedges weight-function models; `puniform` and `puniforms` for p-uniform corrections; `RoBMA` for Bayesian model-averaged publication-bias adjustment; `metasens` for limit-meta-analysis and Copas-model alternatives.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Systematic reviews and PRISMA
- Meta-analysis basics
- Network meta-analysis